NewsBot Midterm - Reflective Journal

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Team Collaboration Analysis

I completed the NewsBot Intelligence System midterm independently rather than working with a group. This meant I was responsible for organizing the project, completing each module, troubleshooting errors, and making decisions about the final system. Although working individually allowed me to maintain control over my progress and decisions, it also meant that I had to manage every challenge myself. One of the biggest lessons was understanding how important it is to plan ahead when different parts of a project depend on one another.

Technical Integration Challenges

The most challenging part of this project was getting the different NLP components to work together correctly. Each module introduced another technique, including text preprocessing, TF-IDF, part-of-speech tagging, syntactic analysis, sentiment analysis, and named entity recognition. Understanding these techniques individually was one thing, but integrating them into a functioning system required additional troubleshooting.

One major challenge involved the dataset. The original notebook contained only ten sample articles, which was not sufficient for the project requirements or meaningful machine learning evaluation. I replaced those samples with the BBC News dataset, containing 1,490 articles across five categories. After removing 50 duplicates, I worked with 1,440 articles.

Changing the dataset also required updating the preprocessing pipeline, classification features, and model evaluation. Another important issue involved ensuring that the final NewsBot used the same TF-IDF features as the trained classifier. Fixing this helped the system classify new articles without feature mismatches.

Business Value Assessment

I can see several practical applications for a system like NewsBot. News organizations and businesses process large amounts of written information, and manually organizing every article would be time-consuming. An automated classification system could help organize content into categories, improve search capabilities, and support personalized news recommendations.

The sentiment analysis and named entity recognition features could also provide useful information about how organizations, individuals, and events are discussed in the news. For example, a business could use similar technology to monitor industry coverage or identify frequently mentioned competitors.

However, I also recognize that automated analysis should not replace human judgment. During testing, some articles received relatively low classification confidence, and sentiment analysis did not always reflect the actual tone of the article. These limitations would need to be considered before using a similar system in a professional environment.

Individual Contributions

Since I completed the project independently, I was responsible for the overall development and integration of the NewsBot system. This included preparing the dataset, applying preprocessing

techniques, analyzing linguistic patterns, training and comparing classification models, examining sentiment and named entities, and testing the final integrated system.

The project achieved 94.79% test accuracy using Naive Bayes, while Logistic Regression achieved a slightly higher test accuracy of 95.49%. Naive Bayes was selected because it had the strongest cross-validation performance. I also evaluated the system using new sample articles and examined where its predictions and extracted information could be improved.

Future Enhancements

If I continued developing NewsBot, I would want to improve its ability to analyze articles that overlap multiple categories. I would also explore more advanced sentiment analysis methods because the current approach sometimes struggles with context and tone.

Another improvement would involve using newer and more diverse news articles to evaluate how well the system performs outside the BBC dataset. I would also like to improve named entity recognition and eventually develop a simple interface where users could enter an article and receive its classification, sentiment, and important entities without needing to interact directly with the code.

Professional Development Impact

This project helped me better understand how different NLP methods connect to create a functioning artificial intelligence system. Rather than only practicing individual techniques, I had to consider how preprocessing decisions affected later analysis and how the machine learning model interacted with the final application.

I also gained more experience troubleshooting code, evaluating model performance, and recognizing that a successful program is not necessarily a perfect one. The system produced strong classification results, but testing revealed limitations that would still matter in real-world use.

As I continue working toward my degree in Robotics and Artificial Intelligence, projects like NewsBot help me build practical experience and strengthen my portfolio. More importantly, this project showed me how much work goes into developing an AI system beyond simply training a model.